

In this project we've used a combination of design patterns. The singleton pattern was used to create the main classes and actors of the project, these include the Device, Headset and PC. Given the persistent aspect of each actor in the scenarios, it made sense to create a single instance that persists throughout the lifetime of the application. The Observer pattern is very prominent in the Qt framework, the UI of the application relies on the signals to be handled by slot functions connected to them by the MainWindow. As such, we attempt to leverage this design pattern where possible to delegate the concurrency of certain operations, where QThreads might have been difficult to manage.